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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CS3314-1 – SOFTWARE TESTING
(For CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. What is meant by 'exhaustive testing'?
A) Testing only critical modules
B) Testing all possible combinations of inputs and conditions
C) Testing only functional requirements
D) Running a single test repeatedly
2. Validation ensures:
A) The system meets user needs and requirements
B) Only documents are verified
C) The system is built as per design
D) Cost of project is reduced
3. In software testing, what is a 'failure'?
A) A defect in code
B) An error in design
C) Software behaving unexpectedly during execution
D) The cost of fixing a bug
4. Which testing principle highlights that the same set of test cases will stop finding new bugs over time?
A) Defect clustering
B) Pesticide paradox
C) Exhaustive testing is impossible
D) Early testing
5. In STLC, which is the first phase?
A) Test Execution
B) Test Closure
C) Test Environment Setup
D) Requirement Analysis
6. Which testing is performed by developers using stubs and drivers?
A) System testing
B) Acceptance testing
C) Unit (component) testing
D) Integration testing
7. What is the main purpose of Integration Testing?
A) To check communication between modules
B) To verify only business requirements
C) To validate documentation
D) To check system speed
8. Boundary value analysis belongs to which test design technique?
A) Black-box
B) White-box
C) Static
D) Dynamic
9. Which of the following best defines Test Management?
A) A way to manage only test execution
B) A process of planning, execution, monitoring, and control of testing activities
C) A tool for bug reporting
D) A document for test data storage
10. Which of the following is NOT usually part of a test plan?
A) Scope and out-of-scope items
B) Test team allocation
C) Source code changes
D) Test schedule
11. Risk analysis in test management is done to:
A) Reduce cost only
B) Detect potential project risks early
C) Increase code size
D) Document test data



12. Issue management deals with:
 A) Avoiding project risks when they occur
 B) Creating test data
 C) Designing UIs
 D) Reducing function points
13. Which tool is used for test case management as a Bugzilla extension?
 A) JIRA
 B) QTP
 C) Selenium
 D) Testopia
14. Which is NOT a goal of defect management?
 A) Prevent defects
 B) Early detection
 C) Minimize impact
 D) Increase complexity
15. Which stage comes first in Test Data Setup?
 A) Identify data needs
 B) Data creation
 C) Data masking
 D) Data cleanup
16. Which describes synthetic test data?
 A) Data generated artificially for testing
 B) Data copied from live system
 C) Data deleted after use
 D) Data migrated from production
17. What is the primary goal of automation testing?
 A) Eliminate manual testing completely
 B) Reduce manual effort by automating repetitive tasks
 C) Increase manual intervention
 D) Only test new features
18. Which tool uses proprietary ML models to repair UI tests automatically?
 A) Rational Functional Tester
 B) Watir
 C) mabl
 D) silktest
19. Which Selenium component allows parallel execution of tests?
 A) Selenium IDE
 B) Selenium RC
 C) Selenium WebDriver
 D) Selenium Grid
20. In Selenium IDE, 'pause' command is used for:
 A) Debugging test steps
 B) Waiting for a specified time
 C) Asserting values
 D) Skipping execution

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is software testing? Discuss the role of a tester to maintain software quality.	8	L*2	1	1
	b) Explain the V-Model of Software Testing Life Cycle (STLC) with its phases.	8	L2	2	1
2.	a) Explain Usability and Adhoc testing techniques in detail.	8	L2	1	1
	b) Outline structure-based testing and specification-based testing with its types.	8	L2	2	2
3.	a) Compare and contrast the different levels of testing.	8	L2	1	1
	b) Explain static techniques in software testing. How do walkthroughs and Inspection improve software quality?	8	L2	2	2
Unit – II					
4.	a) Define Test Management. Describe the various steps of the Test Management process in detail and illustrate the process with a neat diagram.	8	L2	3	1
	b) Illustrate the defect life cycle with a neat diagram.	8	L2	4	1
5.	a) Discuss effort estimation techniques used in software testing.	8	L2	3	1
	b) In detail explain TESTOPIA and BUGZILLA tools.	8	L2	4	1

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| 6. a) | Define Configuration Management in the context of software testing. Explain the tasks involved in the Software Configuration Management process in detail. | 8 | L2 | 3 | 1 |
| b) | Identify and explain the critical challenges encountered in test data management. | 8 | L3 | 4 | 2 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Why is automation preferred over manual testing? Discuss the advantages and limitations of automation testing. | 8 | L2 | 5 | 1 |
| b) | Define Selenium IDE. Explain the different types of locators used in Selenium IDE with examples. | 8 | L2 | 5 | 1 |
| 8. a) | What is the TestNG framework? Explain any six TestNG annotations. | 8 | L2 | 5 | 1 |
| b) | Write a short note on | | | | |
| | i) Selenium Grid. | | | | |
| | ii) Maven automation tool. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
